Rice Production Technology 2024

Package of Practices for Kharif Season

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1. Introduction

Rice (Oryza sativa L.) is the staple food crop for more than 65% of the Indian population. India is the second largest producer of rice globally after China. Rice is primarily a kharif season crop grown during the monsoon period from June to November. The major rice growing states include West Bengal, Uttar Pradesh, Punjab, Tamil Nadu, Andhra Pradesh, Bihar, and Odisha.

1.1 Recommended Varieties

Selection of appropriate variety is the key to successful rice cultivation. The following high yielding varieties are recommended for different regions:

- **Pusa Basmati 1121**: Premium basmati variety for North India with excellent grain quality and aroma. Grain length 8.3 mm, yield potential 40-45 q/ha. Recommended for Punjab, Haryana, and western Uttar Pradesh.
- **Pusa-44**: High yielding non-basmati variety with yield potential of 55-65 q/ha under irrigated conditions. Long duration variety (155-160 days). Suitable for Punjab and Haryana.
- **PR-126**: Short duration variety (123 days) recommended for Punjab. Yield potential 35-38 q/ha. Ideal for rice-wheat rotation to allow timely wheat sowing.
- **Swarna (MTU-7029)**: Most widely cultivated variety in eastern India. Semi-tall with yield potential of 50-55 q/ha. Recommended for Bihar, West Bengal, Odisha, and Chhattisgarh.

2. Nursery Management and Transplanting

2.1 Nursery Raising

Nursery beds should be prepared in the month of May-June depending on the variety and region. The seed rate for transplanted rice is 20 to 25 kg per hectare. For the System of Rice Intensification (SRI), the seed rate is reduced to 5-7 kg per hectare. Nursery area should be one-twentieth of the main field area.

Treat the seed with Carbendazim 50WP at 2 g per kg seed for protection against seed-borne diseases. Soak seed in water for 24 hours and incubate for another 24 hours for uniform germination. Apply 1 kg each of N and P2O5 per 100 sq m nursery area.

2.2 Transplanting

Transplanting should be done when seedlings are 20 to 25 days old (3-4 leaf stage). For SRI method, younger seedlings of 12-14 days with 2 leaves are used. The recommended spacing for transplanting is 20 x 15 cm (row to row x plant to plant) for high yielding varieties. For basmati varieties, use 20 x 20 cm spacing. Plant 2-3 seedlings per hill at a depth of 2-3 cm in the puddled field.

2.3 Soil Requirements

Rice grows well in puddled clay to clay loam soils with good water holding capacity. The ideal soil pH for rice cultivation is 5.5 to 6.5. Rice is also tolerant of slightly acidic conditions unlike wheat. Puddling of the field with 2-3 passes of a puddler in standing water helps in weed control, reduces percolation losses, and creates an anaerobic environment favorable for rice growth.

3. Fertilizer Management

The recommended fertilizer dose for irrigated transplanted rice is 100:50:50 kg N:P:K per hectare. Apply the full dose of phosphorus (50 kg P2O5/ha) and potassium (50 kg K2O/ha) as basal application at the time of last puddling before transplanting.

Nitrogen should be applied in three splits: one-third (33 kg N/ha) as basal at transplanting, one-third at active tillering stage (25-30 days after transplanting), and the remaining one-third at panicle initiation stage (55-60 days after transplanting). Use the Leaf Color Chart (LCC) for need-based nitrogen management to improve nitrogen use efficiency.

Green manuring with Dhaincha (Sesbania aculeata) sown 60 days before transplanting and incorporated at flowering can substitute 25-30 kg nitrogen per hectare. Apply zinc sulphate at 25 kg/ha for zinc-deficient soils, particularly in alkaline and calcareous soils.

4. Irrigation Management

Maintain 5 cm standing water in the field during the vegetative phase of the rice crop from transplanting to tillering. During the reproductive phase, maintain 2-3 cm water. The Alternate Wetting and Drying (AWD) technique can save 20-25% irrigation water without yield loss. In AWD, allow the field to dry until hairline cracks appear before re-irrigating.

Critical stages for irrigation in rice are: transplanting, active tillering, panicle initiation, flowering, and grain filling. Drain the field 10-15 days before harvest to facilitate uniform maturity and easy harvesting. Continuous flooding should be avoided as it leads to higher methane emissions and water wastage.

5. Pest and Disease Management

5.1 Major Insect Pests

• **Stem Borer (Scirpophaga incertulas)**: Causes dead hearts in vegetative stage and white ears in reproductive stage. Apply Cartap hydrochloride 4G granules at the rate of 25 kg per hectare in standing water at 30 days after transplanting. Alternatively, spray Chlorantraniliprole 18.5SC at 0.3 ml per litre.

• **Brown Plant Hopper (BPH)**: Causes hopper burn at the base of tillers. Avoid excess nitrogen application which increases BPH incidence. Spray Imidacloprid 17.8SL at 0.5 ml per litre of water directed at the base of plants. Drain excess water before spraying.

5.2 Major Diseases

• **Blast (Magnaporthe oryzae)**: Causes diamond-shaped lesions on leaves and neck rot. Spray Tricyclazole 75WP at the rate of 0.6 g per litre of water at tillering and boot stage. Avoid excess nitrogen and maintain optimum plant spacing.

• **Sheath Blight (Rhizoctonia solani)**: Causes oval to irregular lesions on leaf sheaths near water level. Spray Hexaconazole 5EC at 1 ml per litre of water or Validamycin 3L at 2 ml per litre. Reduce nitrogen dose in endemic areas.

• **Bacterial Leaf Blight (Xanthomonas oryzae)**: Causes yellow to white lesions from leaf tips. No effective chemical control; use resistant varieties. Avoid excess nitrogen and drain excess water.

6. Expected Yield

Under recommended package of practices, the expected yield of high yielding rice varieties is 55 to 65 quintals per hectare for Pusa-44 under irrigated conditions in Punjab and Haryana. Basmati varieties yield 40-45 q/ha. Short duration varieties like PR-126 yield 35-38 q/ha. In eastern India, Swarna yields 50-55 q/ha under good management. Average national productivity of rice is about 26 q/ha, indicating significant scope for yield improvement through adoption of recommended technologies and integrated crop management practices.